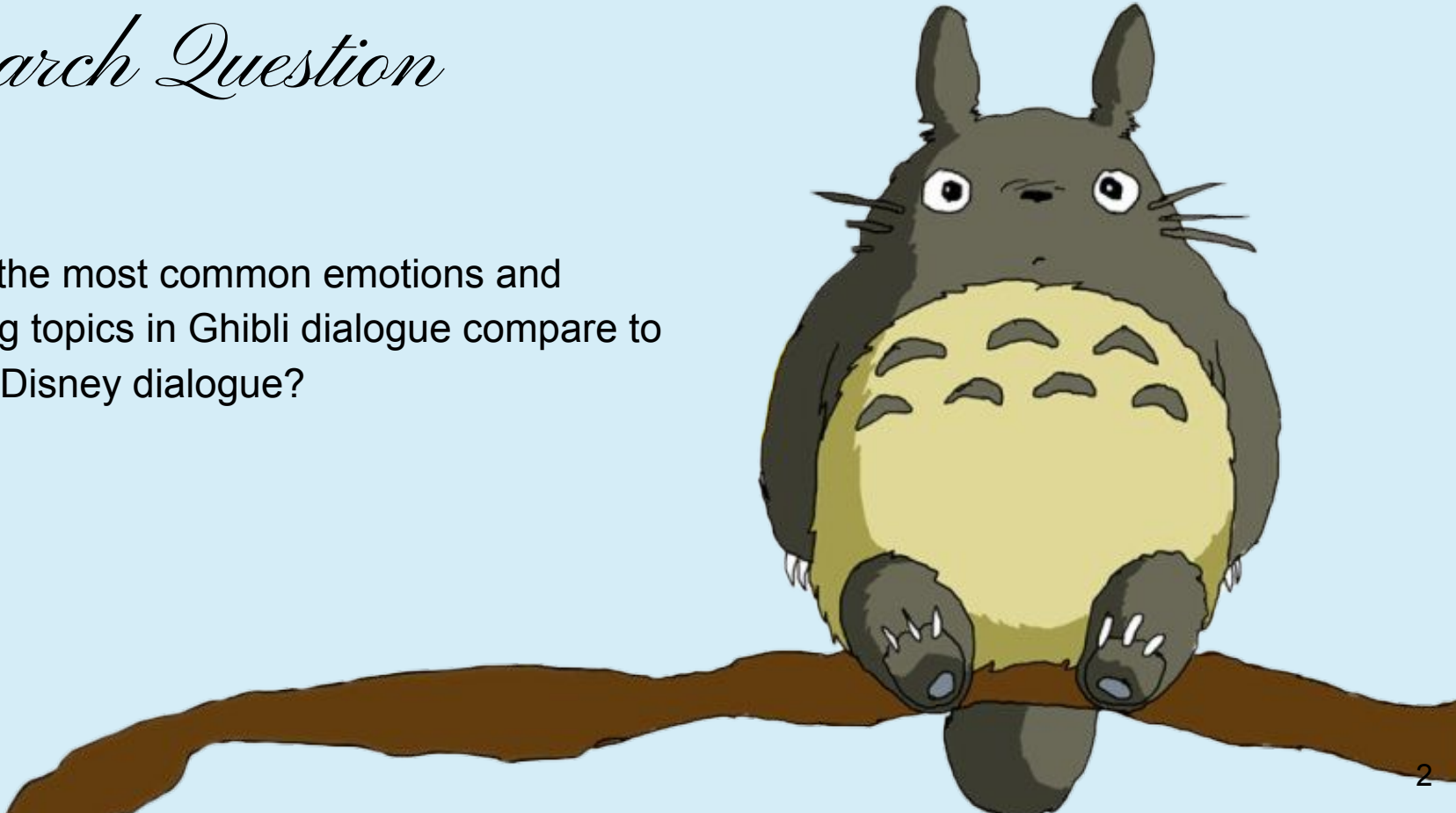


A Comparative Analysis Using Text Mining Techniques



Research Question

How do the most common emotions and prevailing topics in Ghibli dialogue compare to those in Disney dialogue?





Motivation

DATA



```
1      1
2      00:00:54,304 --> 00:00:55,848
3      In the beginning...
4
5      2
6      00:00:56,014 --> 00:00:58,851
7      there was only ocean...
8
9      3
10     00:00:59,017 --> 00:01:02,980
11     until the mother island emerged.
12
13     4
14     00:01:03,146 --> 00:01:04,982
15     Te Fiti.
16
17     5
18     00:01:05,607 --> 00:01:09,361
19     Her heart held the greatest power
20     ever known.
```

Subtitles

- 17 movies from Disney and 17 movies from Studio Ghibli
- Datasource: Just Subtitles

<https://www.justsubtitles.com/>

- Why subtitles?



```
1      1
2      00:00:54,304 --> 00:00:55,848
3      In the beginning...
4
5      2
6      00:00:56,014 --> 00:00:58,851
7      there was only ocean...
8
9      3
10     00:00:59,017 --> 00:01:02,980
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13     4
14     00:01:03,146 --> 00:01:04,982
15     Te Fiti.
16
17     5
18     00:01:05,607 --> 00:01:09,361
19     Her heart held the greatest power
20     ever known.
```

Preprocessing steps

- Leading dashes: dialogue
- HTML tags: strip `<i>...</i>`
- Non-speech cues: all-caps like (LAUGHING) or [MUSIC]
- Speaker labels: deletes e.g. DUMBO: to avoid them in word counts
- Skip logic: filter out pure indices ("1"), timestamps, and blank lines
- Invisible chars



data2 > disney > ≡ Moana.srt

1 1
2 00:00:54,304 --> 00:00:55,848
3 In the beginning...
4
5 2
6 00:00:56,014 --> 00:00:58,851
7 there was only ocean...
8
9 3
10 00:00:59,017 --> 00:01:02,980
11 until the mother island emerged.
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13 4
14 00:01:03,146 --> 00:01:04,982
15 Te Fiti.
16
17 5
18 00:01:05,607 --> 00:01:09,361
19 Her heart held the greatest power
20 ever known.

data_cleaned > disney > ≡ Moana_cleaned.txt

1 In the beginning...
2 there was only ocean...
3 until the mother island emerged.
4 Te Fiti.
5 Her heart held the greatest power
6 ever known.
7 It could create life itself.
8 And Te Fiti shared it with the world.
9 But in time...
10 some began to seek Te Fiti's heart.
11 They believed
12 if they could possess it...
13 the great power of creation
14 would be theirs.
15 And one day...
16 the most daring of them all...
17 voyaged across the vast ocean
18 to take it.
19 He was a demigod of the wind and sea.
20 He was a warrior.
21 A trickster.



```
data_in_sentences > disney > ≡ Wreck-It.Ralph_cleaned_sentences.txt
```

```
1   My name's Ralph, and I'm a bad guy.
2   Let's see.
3   I'm nine feet tall.
4   I weigh 643 pounds.
5   Got a little bit of a temper on me.
6   My passion bubbles very near the surface, I guess, not gonna lie.
7   Anyhoo, what else?
8   I'm a wrecker.
9   I wreck things.
10  Professionally.
11  I'm very good at what I do.
12  Probably the best I know.
13  The thing is, fixing is the name of the game.
14  Literally, Fix-It Felix, Jr.
15  So, yeah, naturally, the guy with the name Fix-It Felix is the good guy.
16  He's nice enough as good guys go.
17  Definitely fixes stuff really well.
18  But if you've got a magic hammer from your father, how hard can it be?
```

Sentences

SENTIMENT ANALYSIS

- `(?<=[!?!])\s+` → period, exclamation, question mark followed by space = split
- Keeps punctuation
- Result: splits into sentences → writes one sentence per line

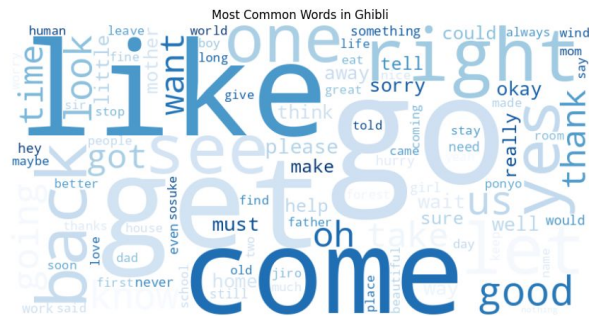
Tokenization

TOPIC MODELING

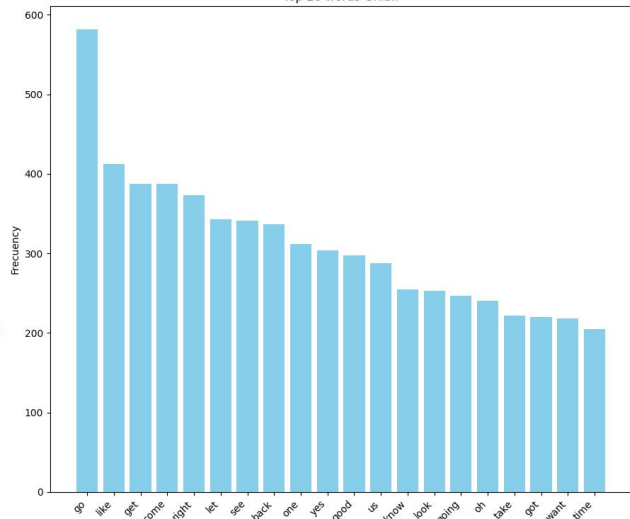
data_preprocessed > disney > ≡ Moana_cleaned_tokens.txt

```
1 beginning ocean mother island emerged te fiti heart held greatest power ever known could create life
te fiti shared world time began seek te fiti heart believed could possess great power creation would
one day daring voyaged across vast ocean take demigod wind sea warrior trickster shapeshifter could
change form power magical fish hook name maui without heart te fiti began crumble giving birth
terrible darkness maui tried escape confronted another sought heart te kā demon earth fire maui
struck sky never seen magical fish hook heart te fiti lost sea even 1 000 years later te kā demons
deep still hunt heart hiding darkness continue spread chasing away fish draining life island island
every one us devoured bloodthirsty jaws inescapable death one day heart found someone journey beyond
reef find maui deliver across great ocean restore te fiti heart save us thank mother enough papa one
goes outside reef safe darkness monsters monsters monsters monsters darkness nothing beyond reef
storms rough seas gonna throw long stay safe island fine legends true someone go mother motunui
paradise would want go anywhere else shoo shoo moana moana scared wanna go back know know go
dangerous moana come let go back village next great chief people wondrous things little minnow oh yes
first must learn meant moana make way make way moana time knew village motunui need dancers
practicing dance ancient song needs new song old one need tradit
need share everything make joke weave baskets fishermen come bac
ground people need chief comes day gonna look around realize hap
use part coconut need make nets fibers water sweet inside use le
consider coconuts trunks leaves island gives us need one leaves
future okay time learn must find happiness right like dance wate
like misbehaves village may think crazy say drift far know like well father daughter stubbornness
pride mind says remember may hear voice inside voice starts whisper follow farthest star moana voice
```

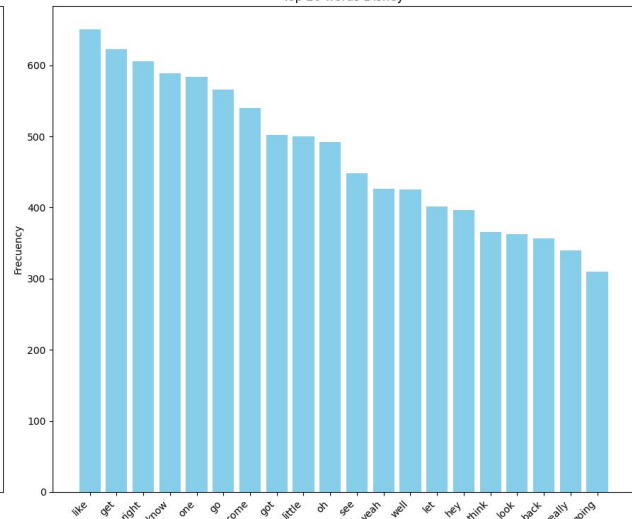
- Remove punctuation
- Lowercase
- Stop words



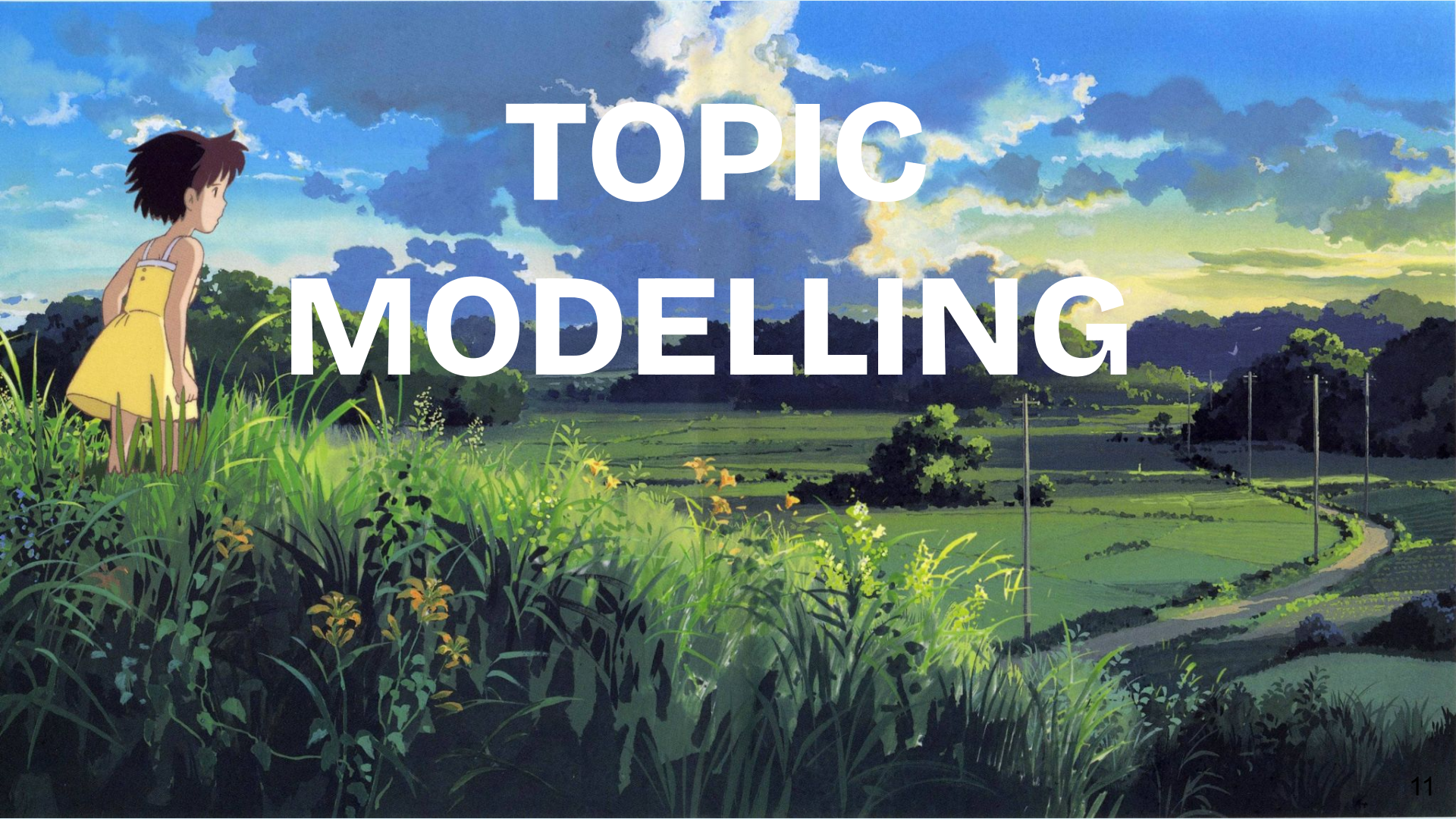
Top 20 words Ghibli



Top 20 words Disney



TOPIC MODELLING

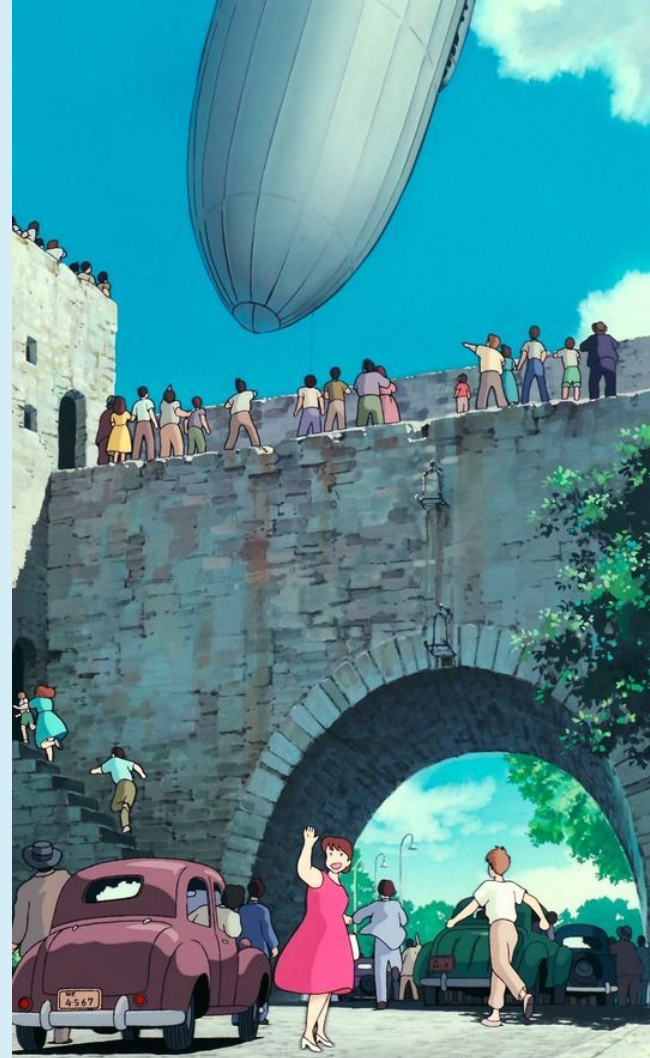


Topic Modelling

We used **Latent Dirichlet Allocation (LDA)** and used to **Coherences Scores** to evaluate the quality of the model

Preprocessing:

- Divided each movie script (17 per genre) in **6 equal part**
→ total document number: **170 for each genre**
- **Part-of-speech tagging** using NLTK's Universal Tagset
- **Lemmatized words** using WordNet Lemmatizer
- Kept only **nouns** to focus on meaningful content words

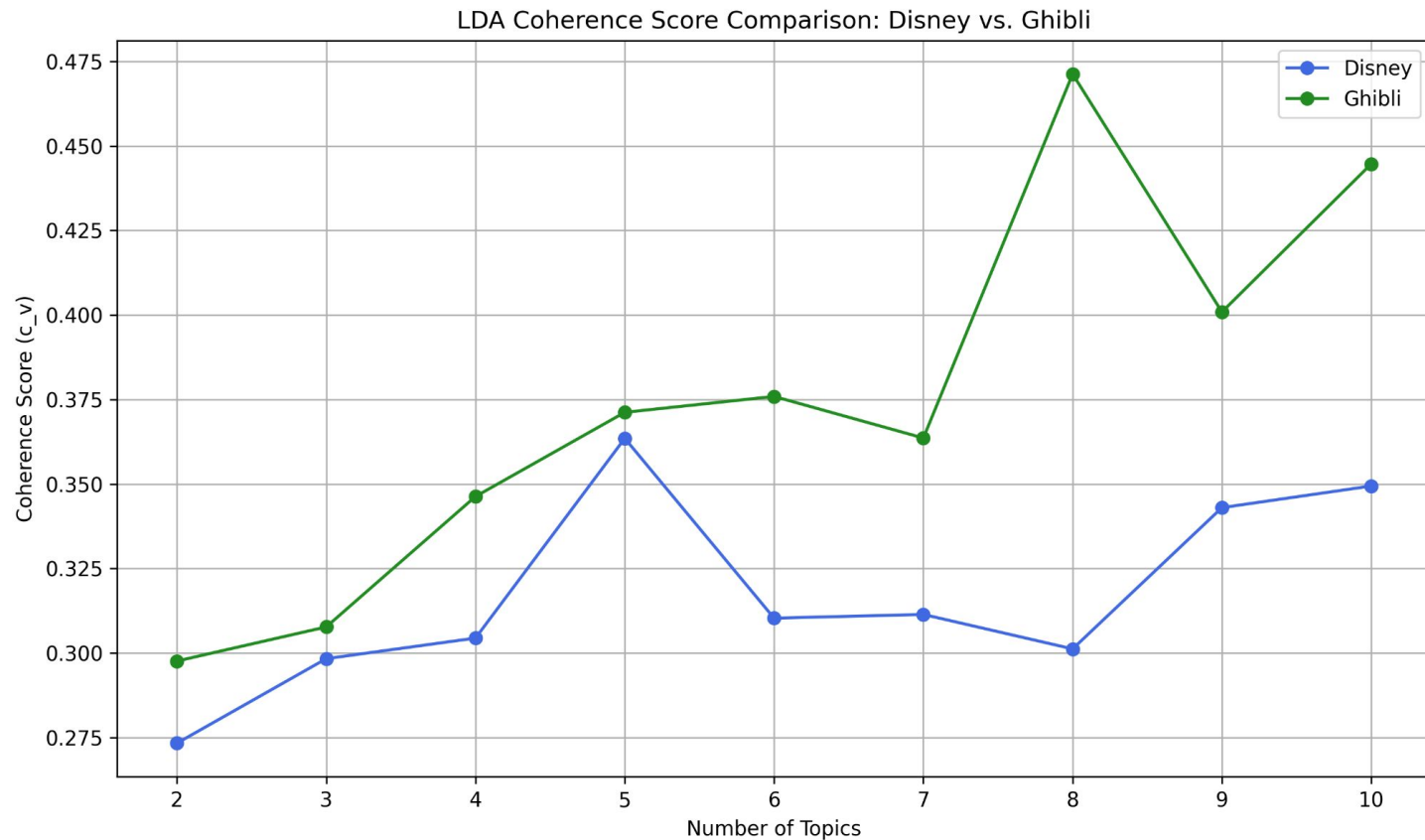


Topic Modelling

- Filtered out **stopwords**: Standard English stopwords (e.g., "the", "is"), Custom stopwords (e.g., character names, studio-specific terms)
- Removed **short tokens** (less than 3 characters)
- Adjusted **hyperparameter**: chunksize and passes
- Filtered tokens by frequency: Removed words that appear in fewer than 3 documents & Removed words that appear in more than 80% of documents



Coherence Scores



Disney:

- 1) **Digital Adventure:** game, kid, friend, **man**, race, home, heart, life
- 2) **The wild world:** heart, night, savage, **predator**, world, **son**, night, rock
- 3) **Dreamed life:** **man**, world, dream, life, heart, talk, home, gold
- 4) **Ocean:** heart, dream, people, island, hook, sea, boat, love
- 5) **Unclear Topic:** **hero**, planet, baby, year, god, place, guest, **prince**

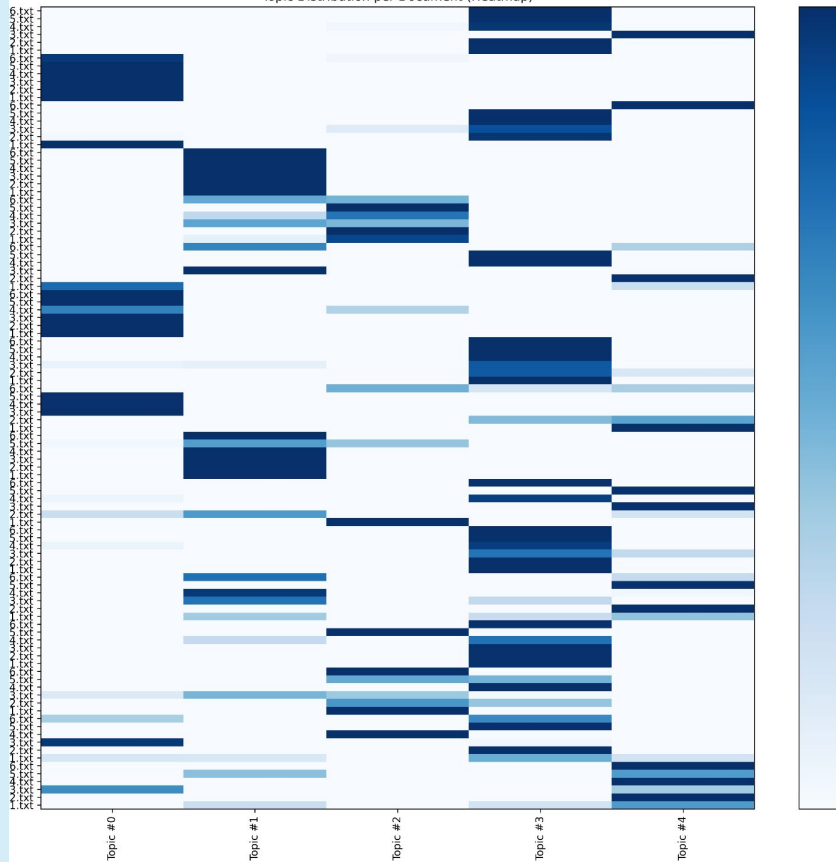
Ghibli:

- 1) **Feminine Power and Nature:** life, spirit, world, lord, **woman**, sea, human, **princess**
- 2) **City life:** home, school, road, hand, people, night, work, town
- 3) **In the air:** plane, engine, air, quarter, club, hand, work, fly
- 4) **Family:** home, **mother**, year, **mama**, house, family, dream, country
- 5) **Magic:** wizard, work, eat, **witch**, morning, dad, home, water



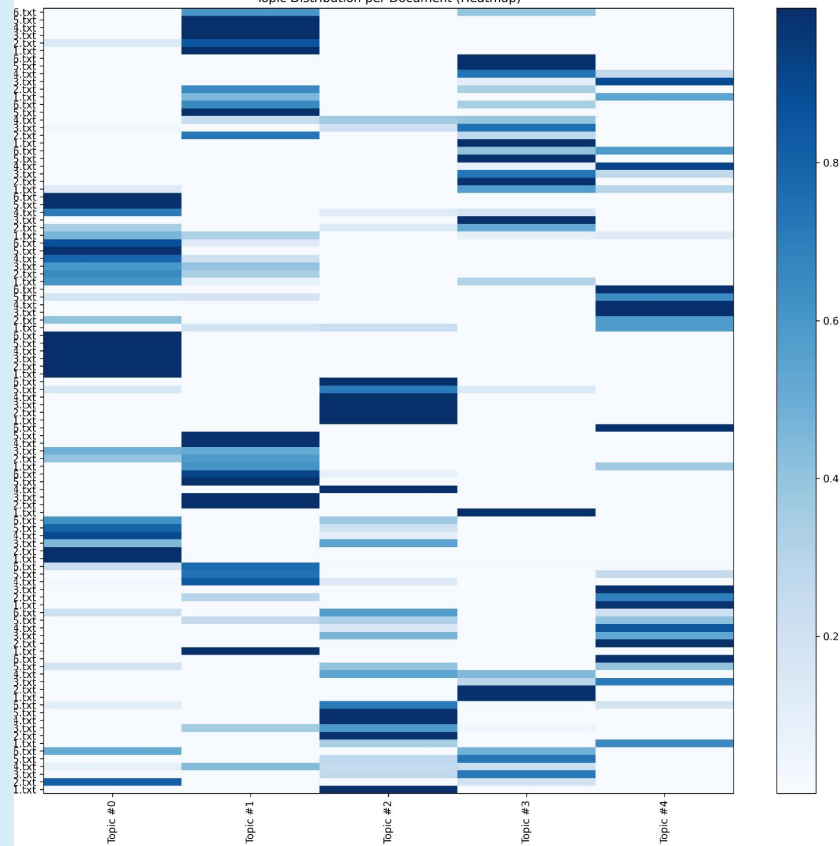
Disney

Topic Distribution per Document (Heatmap)

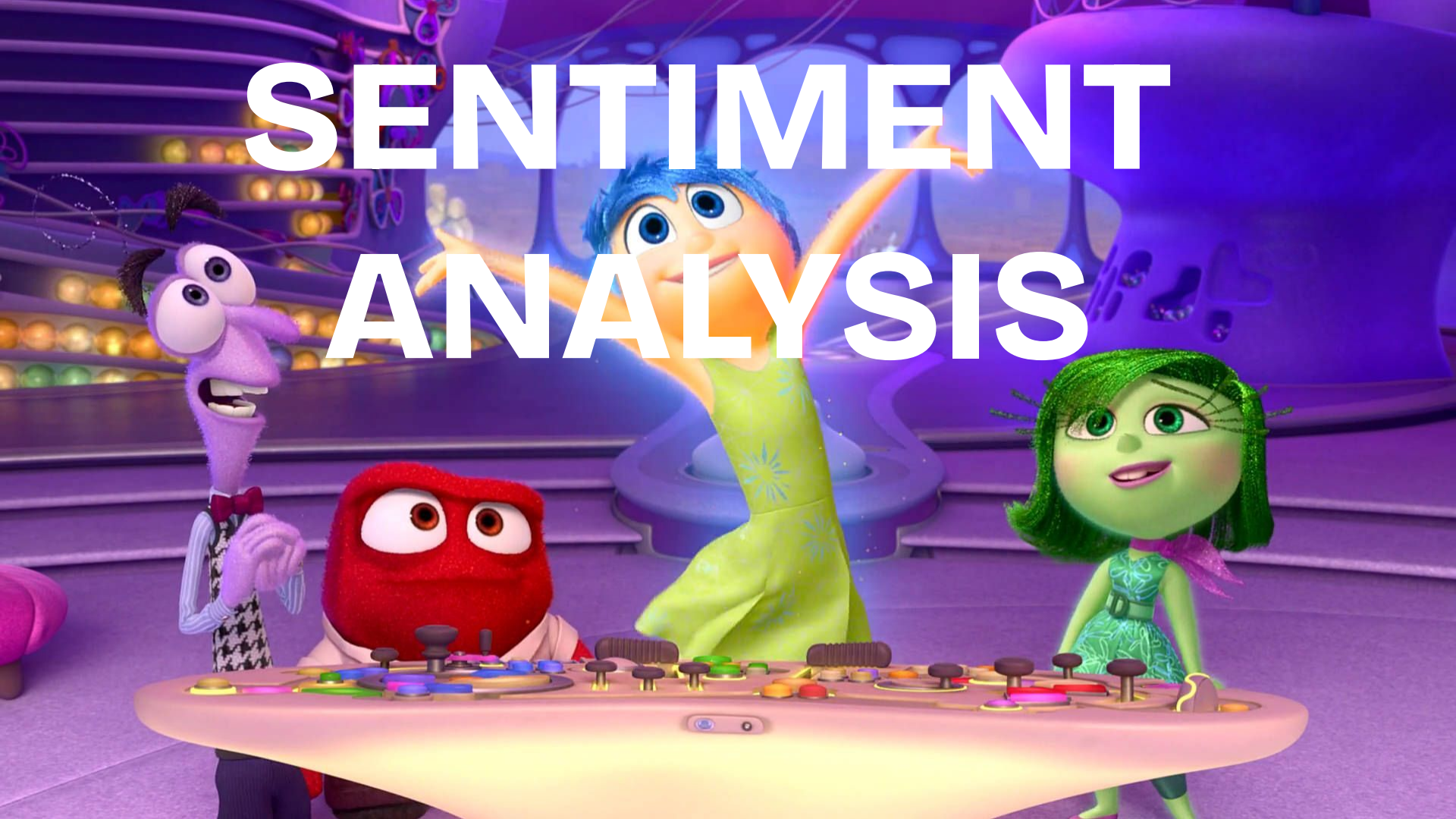


Studio Ghibli

Topic Distribution per Document (Heatmap)



SENTIMENT ANALYSIS



Sentiment Analysis Methods 1

[cardiffnlp/twitter-roberta-base-sentiment](#): fine tuned on millions of labeled tweets

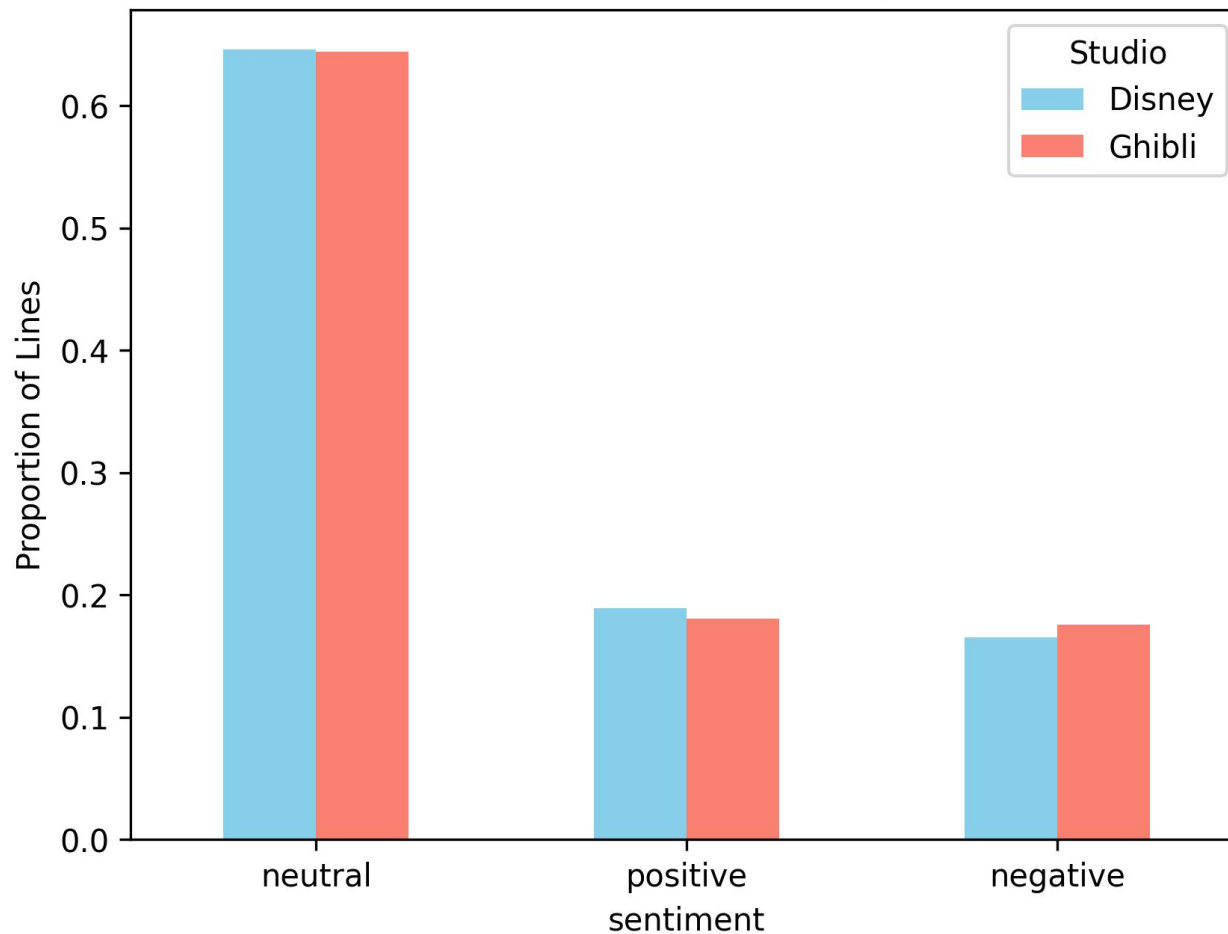
- Tweets are short, informal, and often punctuated like subtitles

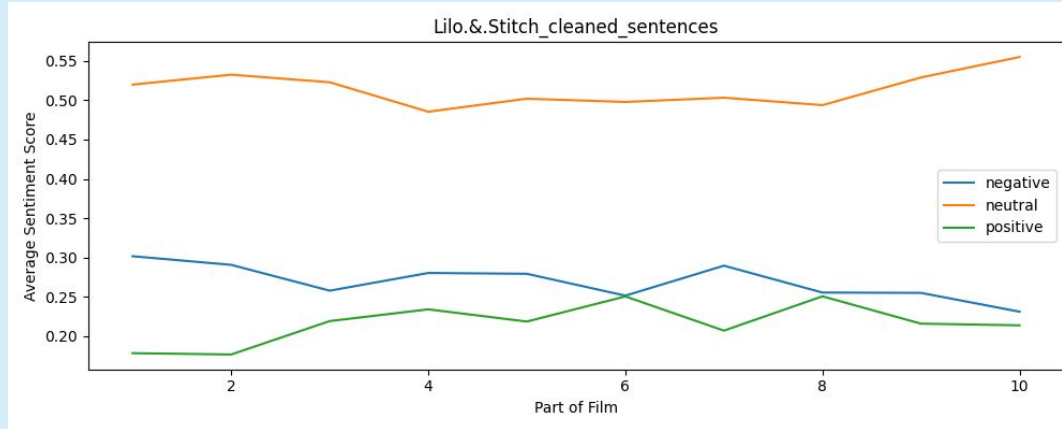
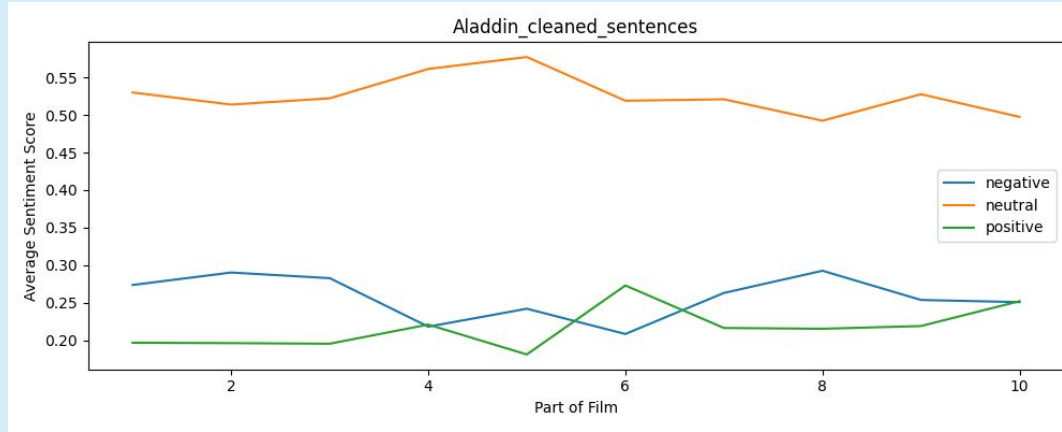
[sarahai/movie-sentiment-analysis](#): positive, neutral, negative, uncertain

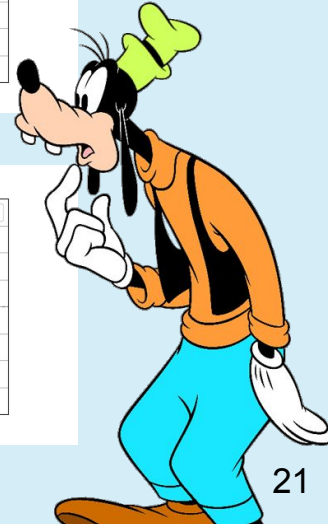
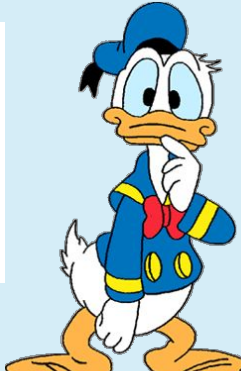
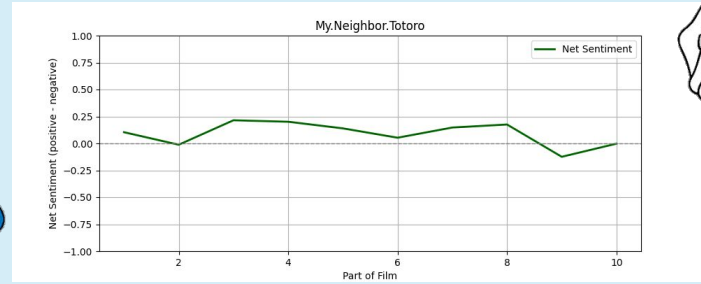
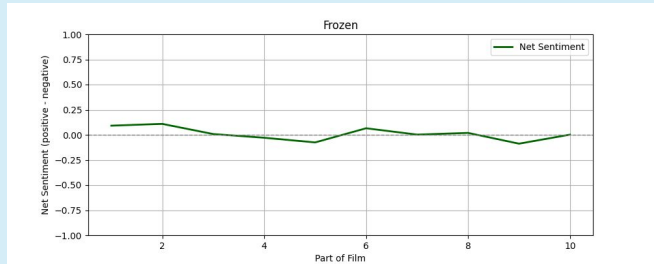
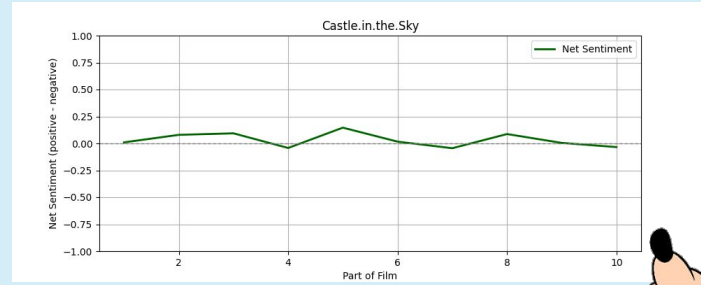
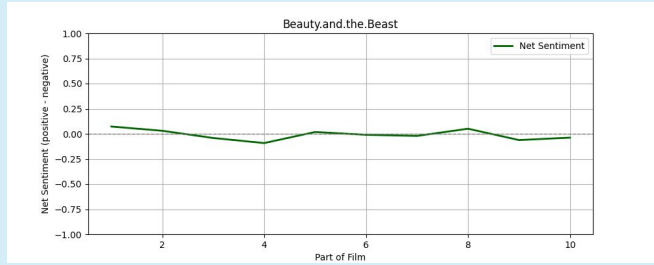
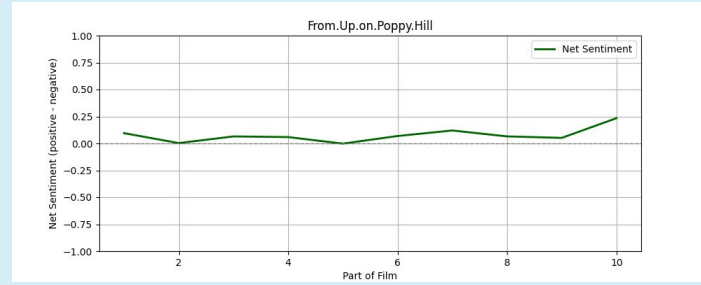
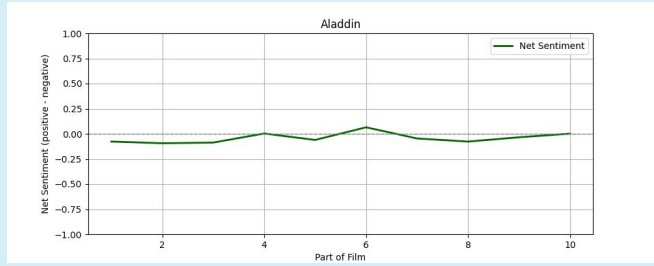
[tabularisai/multilingual-sentiment-analysis](#): very negative, negative, neutral, positive, very positive



Sentiment Distribution Comparison







Sentiment Analysis Method 2

Theme	Ghibli	Disney
Resilience	0.449	0.453
Grief	0.733	0.500
Trauma	0.518	0.510
Personal		
Growth	0.562	0.528
Empathy	0.585	0.505
Romantic Love	0.612	0.497
Friendship	0.850	0.668
Family	0.758	0.691
Loneliness	0.792	0.578
Identity	0.501	0.571

Sentiment Model:

[tabularisai/multilingual-sentiment-analysis](https://huggingface.co/tabularisai/multilingual-sentiment-analysis)

<https://huggingface.co/tabularisai/multilingual-sentiment-analysis>

- Classifies text into 5 categories: very negative, negative, neutral, positive, very positive
- Pre-trained multilingual transformer-based model

Theme Classifier:

[facebook/bart-large-mnli](https://huggingface.co/facebook/bart-large-mnli)

- Zero-shot classification to identify narrative themes
- No specific training required for the defined themes

Limitations

- Limited number of documents
- Difficulty with the models, since they are not trained for movie scripts
- Maybe future labeled data?

Conclusions



WALT DISNEY'S
MICKEY MOUSE

